

Inorganic Chemistry

Inorganic chemistry deals with [synthesis](#) and behavior of [inorganic](#) and [organometallic](#) compounds. This field covers [chemical compounds](#) that are not carbon-based, which are the subjects of [organic chemistry](#). The distinction between the two disciplines is far from absolute, as there is much overlap in the subdiscipline of [organometallic chemistry](#). It has applications in every aspect of the chemical industry, including [catalysis](#), [materials science](#), [pigments](#), [surfactants](#), [coatings](#), [medications](#), [fuels](#), and [agriculture](#).

Occurrence

Many inorganic compounds are found in nature as [minerals](#).^[2] Soil may contain [iron sulfide](#) as [pyrite](#) or calcium sulfate as [gypsum](#).^{[3][4]} Inorganic compounds are also found multitasking as [biomolecules](#): as electrolytes ([sodium chloride](#)), in energy storage ([ATP](#)) or in construction (the [polyphosphate](#) backbone in [DNA](#))

Bonding

[Inorganic compounds](#) exhibit a range of bonding properties. Some are [ionic compounds](#), consisting of very simple [cations](#) and [anions](#) joined by [ionic bonding](#). Examples of salts (which are ionic compounds) are [magnesium chloride](#) MgCl_2 , which consists of [magnesium](#) cations Mg^{2+} and [chloride](#) anions Cl^- ; or [sodium hydroxide](#) NaOH , which consists of [sodium](#) cations Na^+ and [hydroxide](#) anions OH^- . Some inorganic compounds are highly covalent, such as [sulfur dioxide](#) and [iron pentacarbonyl](#). Many inorganic compounds feature [polar covalent](#) bonding, which is a form of bonding intermediate between covalent and ionic bonding. This description applies to many [oxides](#), [carbonates](#), and [halides](#). Many inorganic compounds are characterized by high [melting points](#). Some salts (e.g., [NaCl](#)) are very soluble in water.

When one reactant contains [hydrogen atoms](#), a reaction can take place by exchanging protons in [acid-base chemistry](#). In a more general definition, any chemical species capable of binding to electron pairs is called a [Lewis acid](#); conversely any molecule that tends to donate an electron pair is referred to as a [Lewis base](#).^[5] As a refinement of acid-base interactions, the [HSAB theory](#) takes into account [polarizability](#) and size of ions.

Subdivisions of inorganic chemistry

Subdivisions of inorganic chemistry are numerous, but include:

- [organometallic chemistry](#), compounds with metal-carbon bonds. This area touches on [organic synthesis](#), which employs many organometallic catalysts and reagents.
- [cluster chemistry](#), compounds with several metals bound together with [metal-metal bonds](#) or [bridging ligands](#).
- [bioinorganic chemistry](#), biomolecules that contain metals. This area touches on [medicinal chemistry](#).
- [materials chemistry](#) and [solid state chemistry](#), extended (i.e. polymeric) solids exhibiting properties not seen for simple molecules. Many practical themes are associated with these areas, including [ceramics](#).

Industrial inorganic chemistry

Inorganic chemistry is a highly practical area of science. Traditionally, the scale of a nation's economy could be evaluated by their productivity of [sulfuric acid](#).

An important man-made inorganic compound is [ammonium nitrate](#), used for fertilization. The ammonia is produced through the [Haber process](#).^{[6][7][8]} Nitric acid is prepared from the ammonia by oxidation. Another large-scale inorganic material is [portland cement](#). Inorganic compounds are used as [catalysts](#) such as [vanadium\(V\) oxide](#) for the oxidation of sulfur dioxide and [titanium\(III\) chloride](#) for the [polymerization of alkenes](#). Many inorganic compounds are used as [reagents](#) in [organic chemistry](#) such as [lithium aluminium hydride](#).^[9]